



ENLIGHTENING INDUSTRIES OF ECONOMICS SYSTEM



RISH EM DC

Every system needs proportionate energy from the source for its smooth functioning. Just like Sun Light up the solar system with required amount of energy, RISH EM DC enlightens industries of the economic system with requisite.

RISH EM DC is a compact yet effective DC Energy Meter designed specifically to measure, display and communication DC Voltage, Current, Power and Energy. This enable an industry to monitor and control the target system, making its functions revolve around RISH EM DC

Applications



Wind Power Generation



Solar Photovoltaic Systems



Battery Chargers and Systems



Electroplating Industries



Power Distribution



Measures



Control



Record



Analyse

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nel to be directly replaced by one NB-IoT channel, or many more to be accommodated within an LTE channel where it can share capacity and mobile-site resources with regular LTE data traffic, thus enabling LTE networks to accommodate many more MTC devices.

Mission-critical MTC

At the opposite end of the scale, mission-critical MTC applications include driverless vehicles and robot surgery, and will have stringent requirements on both reliability and low latency. These form one of the key use cases for 5G, and new network strategies including mobile edge computing (MEC) are being developed to cater to these.

The accelerating development of 5G is partly being driven by a demand for expanding the networks to enable the secritical IoT use cases, where the need for low-latency and high-reliability devices cannot be met using existing connectivity technologies.

Looking to the future, 4G, 5G and legacy mobile will all form part of the eventual IoT mix. Many IoT devices need to operate indoors, so issues of in-building penetration need to be addressed as well. New architectures and frequency bands may be needed to improve indoor coverage. A mixture of heterogeneous networks (HetNets) using small cells, Wi-Fi offload and more innovative indoor coverage solutions such as intelligent digital DAS (iDDAS), which allows mobile operators to use flexible repeater solutions to dynamically allocate capacity within buildings or areas is expected to form part of the solution.

Software-defined networks utilising network function virtualisation and self-optimising networks will increasingly be used by operators to help make networks more flexible and responsive to changes in demand. These will form an integral part of the development of the IoT, and in managing network resources

to cope with the extra traffic that it will generate.

Network test and validation

These large volumes and diversity of devices and applications will introduce a new set of challenges in testing and validation, not only for the devices themselves but also for the networks that will support these. Mobile networks that are being expanded to include the IoT will also need new kinds of testing, and the effect of loading the network with millions of these devices will need to be validated.

Although individual devices may not need much bandwidth, with millions more devices coming online, efficiencies of 4G and 5G will be required, in order to increase the overall capacity of the network to be able to handle these.

Availability of network validation tools that emulate realistic data usage scenarios has been vital to the development and rollout of LTE, and in the future it will be equally important to include the characteristics of the predicted volume of IoT devices in the traffic models, in order to ensure quality of service and quality of user experience as 5G networks develop.

The philosophy of testing by design will provide the wireless industry with the blueprint to begin developing this framework, and this will necessitate the use of a validation environment based on virtualised network functions that can apply realistic scenarios for all types of users—consumers, businesses and things.

Latest generation of TM500 network test system already has the capability to emulate certain types of M2M/MTC devices, to evaluate the effect on the network of connecting large numbers of these low-power, low-complexity devices. This helps network operators and infrastructure vendors to validate 4G mobile network performance in preparation for deploying IoT connectivity. **EFY**